Web Endpoint API Getting Started Guide

For ACC™ Software version 7.12 and later

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PDF-WebAPI-C

Revision: 1 - EN

20220202

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What is the Avigilon Control Center™ Software?

The Avigilon Control Center (ACC) video management software allows you to record and view live video from multiple surveillance cameras, monitor alarms, control cameras, and archive video. Surveillance and security teams can use the ACC software for live monitoring and use the archived footage for investigative and forensic purposes. The ACC software gives users control over their workstation with a simplified screen layout, intuitive controls, and customizable features that enable multiple user interactions.

The ACC Server software is the application that captures and records surveillance data from network cameras and encoders. The captured data is sent to the ACC Client software for operators to review. The ACC Client software works with the ACC Server software to give you access and control of your surveillance system. The ACC Client software allows you to view live and recorded video, monitor events, and control user access to the ACC system. The Client software also gives you the ability to configure your surveillance system.

Editions

Note: Features accessed by the ACC Web Endpoint API may not be available on all software editions. For more information about supported features, see the Editions tab on **avigilon.com/acc**.

There are three editions of the ACC software available: Core, Standard, and Enterprise.

The edition determines how many cameras can be connected to the system and the number of simultaneous ACC Client connections and which features are available in the ACC Client software.

Each edition is described below.

Edition	Description
Core	An entry-level software that delivers advanced high-definition surveillance capabilities and ease-of-use for smaller businesses. Supports up to 24 cameras per server.
Standard	A cost-effective security solution that meets the vital and focused needs of mid-size operations. Supports up to 75 cameras per server.
Enterprise	A powerful, open platform for large-scale surveillance needs. Supports an unlimited number of cameras and servers.

These editions can be enabled or disabled based on the license applied. Third-party system integrations are available for the Standard and Enterprise editions.

Sites and Servers

In the ACC system, servers and connected devices are organized into clusters called sites. By organizing the system into sites, you are able to control user access and system-wide events through the site settings. The site settings are stored on the server, or across all servers in a multi-server system.

Within a site, individual servers are responsible for managing the devices that are connected to it. Specifically, the server controls video recording. Using the server settings, you can control when video is recorded, how long it is stored, and how much bandwidth is used to stream video.

Depending on your system and license edition, you may have multiple servers in a site. When there are multiple servers in a site, the site is able to distribute tasks and system data between the servers so that the system can continue running even if a server fails.

Events and Alarms

Events are occurrences that are logged by the ACC software and may or may not be of interest.

Alarms are events that require an operators attention and are defined by an ACC system administrator.

To ensure important events are reviewed and to minimize false alerts, create specific alarms in the ACC system.

What is the ACC Web Endpoint API?

The ACC Web Endpoint API is a common API to ACC servers for third-party Integrations. It allows interaction with ACC sites, servers, and devices. The API allows third-party applications to stream live and recorded video, read and update alarms, and query events. It runs on both Linux® and Windows operating systems and offers a REST interface.

The ACC Web Endpoint API vs the .NET SDK

The following table describes the ACC Web Endpoint API features compared to the .NET SDK.

Feature		API	SDK
General	Secure connection and login	✔	✔
	Transport layer security Secure Remote Password (TLS-SRP)	—	✔
	Scalable	✔	—
	Platform-independent	✔	—
Video	Get cameras	✔	✔
	View live and recorded video	✔	✔
	Live video with High Definition Stream Management (HDSM) 2.0	—	✔
	Trigger video recording	✔	—
	Display video analytics bounding boxes	✔	✔
	Fisheye dewarping	—	✔
Events and Alarms*	Live event monitoring	✔	✔
	Event query	✔	✔
	Recorded timeline events	✔	—
	Event subscription filters	✔	—
	Live alarm monitoring	✔	✔
	Alarm query	✔	✔
	Alarm actions	✔	✔
Digital inputs and outputs	Trigger digital outputs	✔	✔

Feature		API	SDK
Audio	Audio talkdown**	✔	✔
	Audio listen	✔	✔
PTZ	PTZ control	✔	✔
	PTZ presets	✔	✔
Bookmarks	Add bookmark	✔	✔
	List bookmarks	✔	✔
	Edit bookmarks	✔	✔
	Delete bookmarks	✔	✔
Export	Video export	—	✔
	Audio export	—	✔
	Snapshot export	✔	✔
Configuration	License Plate Recognition (LPR) watch lists	✔	—
	Recording schedules	✔	—

* Events available through the Web Endpoint API include: Camera Health, System Health, Analytics Events, Face Match Events, LPR Events, Alarm Events, and Access Control Events.

** Audio talkdown is only available for windowed streaming in the .NET SDK. Callback streaming does not support audio talkdown.

Web Endpoint API System Requirements

ACC Web Endpoint Service version should match the ACC Server version 7.12 or later.

ACC Server Software Requirements

See system requirements on the Avigilon website: **avigilon.com/products/acc/system-requirements**.

Licensing

A license key is required to activate the ACC Server. The ACC Server has a 30-day trial period. Developers can request licenses by submitting **the request form** for approved developers. No additional licensing is required to use the ACC Web Endpoint API.

Developing with the ACC Web Endpoint API

The ACC Web Endpoint API can be used to integrate your application with the ACC system.

To develop with the ACC Web Endpoint API:

1. Install the ACC Web Endpoint Service.
2. Enable the playground.
3. Authorize your login request.
4. Deploy an integration.

Each step is described in this section.

Installing the ACC Web Endpoint Service

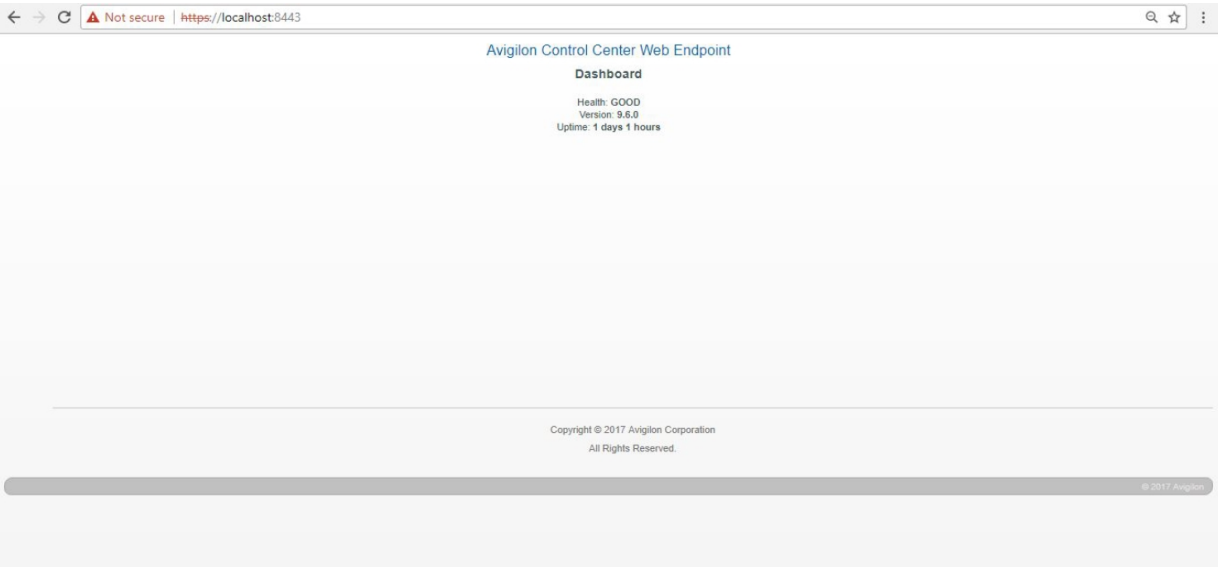
The ACC Web Endpoint Service must be installed on the same system as the ACC Server software.

To install the ACC Web Endpoint Service:

- Download and install the ACC 7 Web Endpoint Service from the Avigilon website: **https://www.avigilon.com/software-downloads**

  Once installed, you can view the health of the ACC 7 Web Endpoint Service at: **https://localhost:8443/**

  Note: On Avigilon appliances, the Web Endpoint Service runs on the standard port 443.



Changing the Default Port

The default port for the ACC Web Endpoint Service is 8443. You can change the default port for integrations on Windows-based systems developed using the ACC Web Endpoint API.

1. In the `%ProgramData%\Avigilon\` folder, open the `WebEndpoint.config.yaml` file in a text editor.
2. Add the following config parameter to the file, where `123` is the new port number:

   ```
   publicRestInterface:
   port: 123
   ```
3. Save the config file and restart the ACC Web Endpoint Service.

   The default port is updated. All commands should be sent to the new port.

Increasing Security

By default, the ACC Web Endpoint has a response header that allows access from any origin. To increase the security of your application, update the YAML configuration file.

1. In the `%ProgramData%\Avigilon\` folder, open the `WebEndpoint.config.yaml` file in a text editor.
2. Add the following config parameter to the file:

   ```
   enableCors: false
   ```
3. Save the config file and restart the ACC Web Endpoint Service.

Enabling the Playground

The ACC Web Endpoint playground provides access to API routes and route documentation on Windows-based systems. To enable the playground, update the YAML configuration file.

Important: The ACC Web Endpoint playground should never be enabled on a production system. It should only be used for development purposes.

1. In the `%ProgramData%\Avigilon\` folder, open the `WebEndpoint.config.yaml` file in a text editor.
2. Add the following config parameter to the file:

   ```
   enablePlaygrounds: true
   ```
3. Save the config file and restart the ACC Web Endpoint Service.

   The ACC Web Endpoint playground is enabled.
4. In a web browser, access the playground: **https://localhost:8443/mt/playground**.

Web Endpoint API Documentation

Route documentation can be found at: **https://localhost:8443/mt/playground/rest**.

Tutorials on the following topics can be found at: **https://localhost:8443/mt/playground**.

- Generating authorization tokens
- Event monitoring with webhooks and socket.io
- Filtering event notifications

Authorization

To use the ACC Web Endpoint Service, you must submit a login request with a valid authorization token.

An authorization token is a string that follows the format:

```
userNonce + ":" + timestamp + ":" + hexEncoded + ":" + integrationIdentifier
```

```
myusernonce:1530726298:89e1dd71c639b78732e2ce7404040de8ec71a7e163833c8efb1c88e
42bc3970e:myIntegrationId
```

Each string component is described below.

String Component	Type	Required	Description	Example
`userNonce`	string	Required	The User Nonce provided by Avigilon.	`myusernonce`
`timestamp`	string	Required	The timestamp in Unix time.	`1530726298`
`hexEncoded`	string	Required	The hex-encoded result of an sha-256 hash of the following combination string: `timestamp + userKey`	`89e1dd71c639b78732e2ce7404040de8ec71a7e163833c8efb1c88e42bc3970e`
`integrationIdentifier`	string	Optional	The integration identifier provided by Avigilon.	`myIntegrationId`

You may request a Web Endpoint API User Nonce and User Key by submitting **this request form**.

The API Playground has a sample Authorization Token Generator that creates an authorization token. You can use this token in playground login requests for up to 1 hour. For more information, see *Authorizing the Playground* below.

To integrate your third-party application with the ACC Web Endpoint Service, you will need to generate an authorization token for each login request. For more information, see *Generating an Authorization Token* on the next page.

> **Note:** The User Nonce and User Key are used for the entire application and should be part of the application code. End users do not need to input their own User Nonce or User Key.

Example

The following example shows a login request and response using a generated authorization token.

Login Request

```
{
       "username": "myUserName",
       "password": "myPassword",
       "authorizationToken":
"myusernonce:1530726298:89e1dd71c639b78732e2ce7404040de8ec71a7e163833c8efb1c88e42bc3970e",
       "clientName": aClientName"
}
```

Login Response

```
{
       "status": "success",
       "result": {
       "session": "AWoAlf8KEPoaF40rb0HvtHxqUP0FYrkSFgoUulTTSKYXPkxHFN18Dup7s1ZrV9kaEGf-
SGiaukGAqurNF_LtNCwiEM3-EY3v90zjqwRoRhDmmosqBnN0cmluZzISGhBNnqyEcsBG_7-IjfJFV8tX",
       "externalUserId": "xddF3Rc-QvKCojvkgf2Wgg",
       "domainId": "-hoXjStvQe-0fGpQ_QViuQ"
}
```

Authorizing the Playground

The token is valid for one hour. Within that hour, you can use the same token for multiple login requests.

To generate an authorization token:

1. In the API Playground, go to the **Authorization Token Generator**.
2. Enter your **User Nonce**, **User Key**, and optional **Integration Identifier**.

   You may request a Web Endpoint API User Nonce and User Key by submitting **this request form**.
3. Click **Generate**.

To authorize your login request:

- Include the generated authorization token in the `authorizationToken` parameter.

Deploying an Integration

To use the Web Endpoint API, you need an HTTP client and an MP4 player. Most browsers and mobile platforms can play the provided video content.

You can deploy an API-based integration using many frameworks and languages, such as hosted web, mobile apps, native Windows/MacOS apps, Electron apps, etc.

Generating an Authorization Token

To integrate your application with the ACC Web Endpoint Service, use the following node.js code to generate an authorization token for each login request. The crypto library is used to encrypt the `timestamp + userKey`.

```
const crypto = require('crypto');

const TokenGenerator = {
        getToken() {

                const timetsamp = (Date.now() / 1000 | 0).toString();
                const hash = crypto.createHash('sha256');

                const userNonce= 'myUserNonce';
                const userKey= 'myUserKey';

                const encrypted = hash.update(`${timestamp}${userKey}`).digest('hex');
                const authToken = `${userNonce}:${timestamp}:${encrypted}`;
                return authToken;
        }
};
```

Include the generated token in the login request.

Sample Application

Sample applications are available on the **Avigilon Developer Forum**.

The following is an example of how you can use the Web Endpoint API to interact with the ACC software.

Demo App for Alarm Management

The ACC Alarm Demo Application is a standalone application designed to demonstrate how to interact with alarms using the ACC Web Endpoint API. The application is written in C# and uses the `System.RunTime.Serialization.json` package to serialize and de-serialize JSON objects to communicate with the ACC Web Endpoint Service.

With the application, you can do the following:

- Perform authentication with the ACC Web Endpoint Service
- Search for alarms
- Monitor alarm status
- Interact with an alarm from a third-party system

Use Cases

Gathering Alarm Information

Gather alarm information for a particular site, including:

- Which alarms are configured on the site
- Current alarm status
- Alarm details

This information is required for the other two use cases and can be used to validate your alarm configuration after the initial site set-up.

Monitoring Alarms

Monitor the ACC system for alarm state changes from a third-party application. This sample application can also be used as a standalone application.

Responding to Alarms

From a third-party application, you can respond to alarms as follows:

- Acknowledge
- Assign

- Purge
- Trigger
- Un-assign

Technical Support

If you require any assistance with the Web Endpoint API, contact us at **developer.avigilon.com**.